THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 12: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. To be called a quadrilateral ABCD, the four distinct points A, B, C, D must have: 1

(a) points in three different planes
(b) all four points on one line
(c) three points collinear
(d) no three collinear, all in one plane, and non-self-intersecting sides

2. The figure that is a non-convex quadrilateral is the one that: 1

(a) has all angles equal **(b)** has one internal angle greater than 180°
(c) has no diagonals **(d)** has two parallel sides

3. A quadrilateral with one pair of opposite sides equal and parallel is: 1

(a) a kite **(b)** a parallelogram **(c)** a rhombus **(d)** a trapezium only

4. The centroid divides each median in the ratio (vertex part : other part): 1

(a) 2 : 1 **(b)** 3 : 1 **(c)** 1 : 1 **(d)** 1 : 2

5. Which of these can be used to tile the plane? 1

(a) only rectangles **(b)** only squares and rhombuses
(c) only convex quadrilaterals **(d)** any quadrilateral, even a non-convex one

6. **Assertion (A):** In a triangle ABC the segment PQ joining the midpoints of AB and AC has length BC/2. 1
Reason (R): PQ is parallel to BC.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

7. In a parallelogram ABCD, $\angle A = 125°$. Then $\angle B$ is: 1

(a) 55° **(b)** 28° **(c)** 235° **(d)** 125°

8. In $\triangle ABC$, P and Q are the midpoints of AB and AC and BC = 9 cm. The length of PQ is: 1

(a) 4.5 cm **(b)** 6.5 cm **(c)** 18 cm **(d)** 9 cm

9. The median AD of $\triangle ABC$ has length 24 cm and G is the centroid. Then AG is: 1

(a) 8 cm **(b)** 22 cm **(c)** 12 cm **(d)** 16 cm

10. The sides of a triangle are 10 cm, 24 cm and 26 cm. The perimeter of the triangle formed by joining the midpoints of its sides is: 1

(a) 60 cm **(b)** 30 cm **(c)** 58 cm **(d)** 32 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why must no three vertices of a quadrilateral ABCD be collinear? What shape do we get if three are collinear? 2

12. Define a convex quadrilateral. Give one visual test using its diagonals. 2

13. In a parallelogram ABCD, $\angle A = 72^\circ$. Find $\angle B$, $\angle C$, $\angle D$. 2

14. Write the converse of 'the diagonals of a parallelogram bisect each other'. Is it true? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that the opposite sides of a parallelogram are equal. 3

16. Prove that if the opposite sides of a quadrilateral are equal, it is a parallelogram. 3

17. Prove that if the opposite angles of a quadrilateral are equal, it is a parallelogram. 3

OR

Prove that if the diagonals of a quadrilateral bisect each other, it is a parallelogram. 3

18. Prove that a quadrilateral with one pair of opposite sides equal and parallel is a parallelogram. 3

OR

The diagonal AC of parallelogram ABCD bisects $\angle A$. Show that it also bisects $\angle C$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that in a parallelogram 5

(a) opposite sides are equal,

(b) opposite angles are equal,

(c) the diagonals bisect each other.

OR

Prove that a quadrilateral is a parallelogram if 5

(a) its opposite sides are equal,

(b) its diagonals bisect each other.

20. Prove that a quadrilateral is a parallelogram if 5

(a) its opposite angles are equal,

(b) one pair of opposite sides is equal and parallel.

OR

State and prove the Midpoint Theorem. Use it to find PQ in $\triangle ABC$ when P, Q are the midpoints of AB and AC and BC $= 26$ cm. 5

END OF QUESTION PAPER — Total: 40 Marks